Statistical Models, Supplementary Material

Can we prove that the Gibbs algorithm keeps its promise?

Systematic Scan Gibbs Sampler

Consider a multivariate random variable $X = (X_1, \dots, X_p)$, and let F denote its distribution.

If we want to sample from a Markov chain that has F as its stationary distribution, we can use the **Gibbs sampler** introduced in the lecture. There are many variants of this algorithm, we looked at the *systematic scan Gibbs sampler*. As you might remember, this algorithm makes use of the *full conditional* distributions.

To describe these distributions rigorosly, we need some notation:

- let $X_{(-i)}$ denote the vector X without its i-th element:

$$X_{(-i)} := (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_p).$$

- let $f(x_1, \dots, x_p)$ denote the density of the joint distribution of $X_1, \dots, X_p$,
- then the marginal density of $X_{(-i)}$ is obtained by integrating over the i-th component, and we denote it by

$$f_{(-i)}(x_{(-i)}) := \int f(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_p) \mathrm{d}y,$$

  where the integration is over the domain of X_i.

Now the conditional density of X_i given $X_{(-i)} = x_{(-i)}$ is

$$f_i(x_i \mid x_{(-i)}) = f(x)/f_{(-i)}(x_{(-i)}).$$

The densities $f_1, \dots, f_p$ are called the *full conditionals*.

Gibbs sampling consists of generating one component at a time from the full conditionals. A complete update thus consists of p steps.

Algorithm: Systematic Scan Gibbs Sampler

- **Intialization**: Start with $X^{(0)} = (x_1^{(0)}, \dots, x_p^{(0)})$, where $f(x^{(0)}) > 0$ to make sure that the start point $X^{(0)}$ lies within the support of X.
- **Update** Given $X^{(s)} = (x_1^{(s)}, \dots, x_p^{(s)})$, obtain $X^{(s+1)} = (X_1^{(s+1)}, \dots, X_p^{(s+1)})$ step by step, generating

1. $X_1^{(s+1)} \sim f_1(\,.\mid x_2^{(s)}, \ldots, x_p^{(s)})$
2. $X_2^{(s+1)} \sim f_2(\,.\mid x_1^{(s+1)}, x_3^{(s)}, \ldots, x_p^{(s)})$

$\vdots$

p. $X_p^{(s+1)} \sim f_p(\,.\mid x_1^{(s+1)}, \ldots, x_{p-1}^{(s+1)})$

Kernel of the Gibbs sampler

The transition kernel corresponding to a complete update has density

$$p((x_1, \ldots, x_p), (y_1, \ldots, y_p)) = \prod_{i=1}^{p} f_i(y_i \mid y_1, \ldots, y_{i-1}, x_{i+1}, \ldots x_p).$$

The Gibbs sampler does its job

Theorem 0.1. *The joint distribution given by the density f is a stationary distribution of the Markov chain produced by the Systematic Scan Gibbs Sampler.*

Proof. We show that for every single component update leaves the distribution unchanged if it was the stationary distribution before the update. The proposition then follows by "finite induction".

Let

$$X^{(s:k)} := (X_1^{(s+1)}, \ldots, X_{k-1}^{(s+1)}, X_k^{(s)}, \ldots, X_p^{(s)})$$

denote the substate before the k-th single component update.

Note that $X^{(s\,:\,k+1)}$ thus denotes the state after the k-th update, and that $X^{(s\,:\,p+1)} = X^{(s+1)}$.

The k-th update leaves the distribution unchanged iff for all measurable B and all $k = 1, \ldots, p$,

$$X^{(s\,:\,k)} \sim f \implies \mathbf{P}(X^{(s\,:\,k)} \in B) = \mathbf{P}(X^{(s\,:\,k+1)} \in B).$$

Assuming that $X^{(s\,:\,k)} \sim f$, we have

$$\mathbf{P}(X^{(s\,:\,k)} \in B) = \underbrace{\int \ldots \int}_{p \text{ integrals}} \mathbf{1}_B(x_1, \ldots, x_{k-1}, x_k, x_{k+1}, \ldots, x_p) f(x_1, \ldots, x_p) \mathrm{d}x_1, \ldots \mathrm{d}x_p$$

and

$$\mathbf{P}(X^{(s:k+1)} \in B) = \underbrace{\int \dots \int}_{p \text{ integrals}} \int \mathbf{1}_B(x_1, \dots, x_{k-1}, y, x_{k+1}, \dots, x_p) f_k(y \mid x_{(-k)}) \mathrm{d}y f(x_1, \dots, x_p) \mathrm{d}x_1, \dots \mathrm{d}x_p$$

$$= \underbrace{\int \dots \int}_{p+1 \text{ integrals}} \mathbf{1}_B(x_1, \dots, x_{k-1}, y, x_{k+1}, \dots, x_p) f_k(y \mid x_{(-k)})\, f_{(-k)}(x_{(-k)}) f_k(x_k \mid x_{(-k)})$$

$$\times \mathrm{d}y\, \mathrm{d}x_1, \dots \mathrm{d}x_{k-1}, \mathrm{d}x_{k+1}, \dots \mathrm{d}x_p \mathrm{d}x_k$$

$$= \underbrace{\int \dots \int}_{p+1 \text{ integrals}} \mathbf{1}_B(x_1, \dots, x_{k-1}, y, x_{k+1}, \dots, x_p) f(x_1, \dots, x_{k-1}, y, x_{k+1}, \dots, x_p)$$

$$\times \mathrm{d}x_1, \dots \mathrm{d}x_{k-1}, \mathrm{d}y, \mathrm{d}x_{k+1}, \dots \mathrm{d}x_p \underbrace{f_k(x_k \mid x_{(-k)}) \mathrm{d}x_k}_{\text{integrates to 1}}$$

$$= \underbrace{\int \dots \int}_{p \text{ integrals}} \mathbf{1}_B(x_1, \dots, x_{k-1}, x_k, x_{k+1}, \dots, x_p) f(x_1, \dots, x_p) \mathrm{d}x_1, \dots \mathrm{d}x_p.$$

That was it. □